

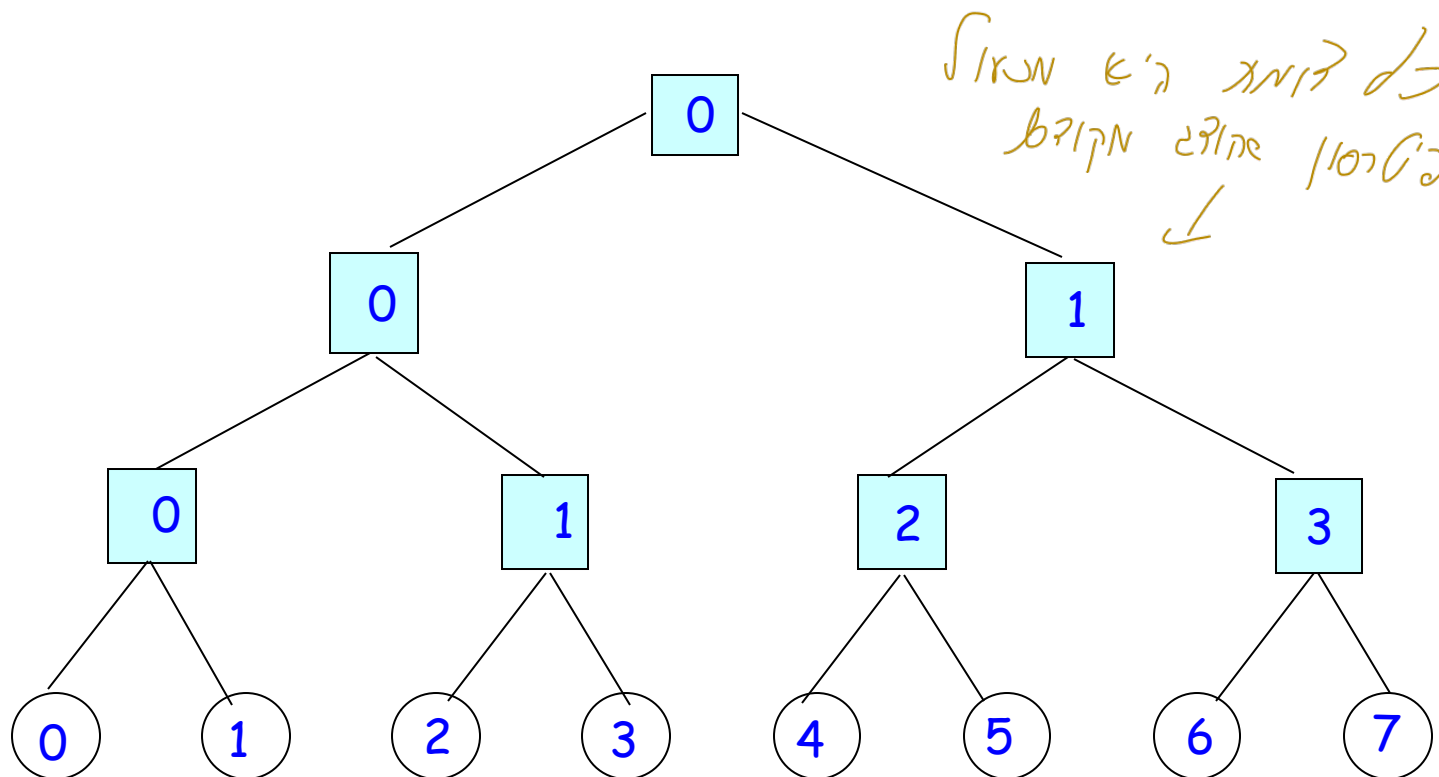
Mutual exclusion for n processes: A tournament tree

Level 2

Level 1

Level 0

Processes



A tree-node is identified by: [level, node#]

N-process mutual exclusion: a tournament tree

Variables

Per node: $b[\text{level}, 2\text{node}]$, $b[\text{level}, 2\text{node}+1]$, $\text{turn}[\text{level}, \text{node}]$

Per process (local): level , node , id .

Program for process i

```
1.  node:=i
2.  For level = 0 to log n-1 do
3.      id:=node mod 2
4.      node:=⌊node/2⌋
5.      b[level, 2node+id]:=true
6.      turn[level, node]:=id
7.      await (b[level, 2node+1-id]=false or turn[level, node]=1-id)
8.  end do
9.  CS
10. for level=log n -1 downto 0 do
11.     node:=⌊ i/2level ⌋
12.     b[level, 2node+id]:=false // id is value set at this node in entry
13. end do
```

